UFW and SSH

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Author Note

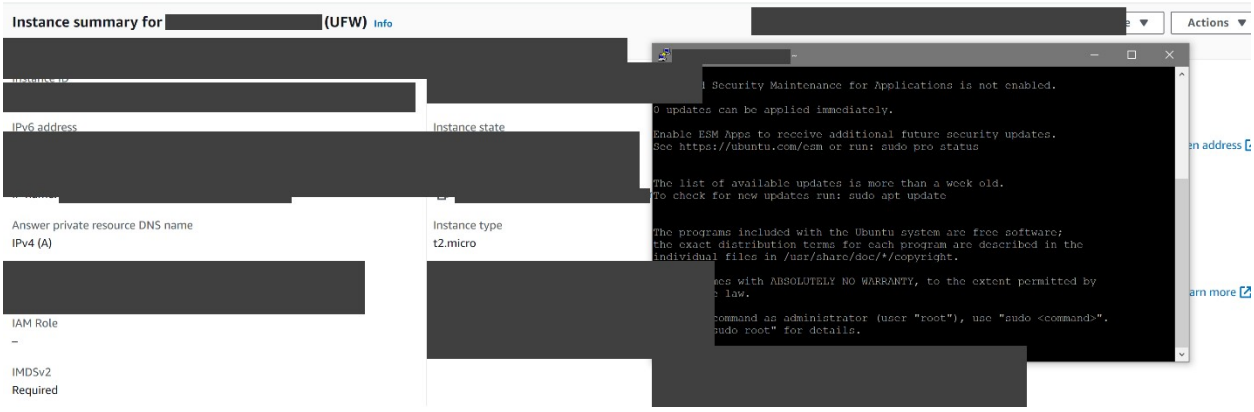
This paper was prepared for NTW216, Foundations of Systems Administration taught by Robert Benton.

UFW and SSH

This paper details the launch of an AWS instance using an Ubuntu operating system. The purpose of the instance is to allow us to gain familiarity with Uncomplicated Firewall (UFW) on Ubuntu servers.

Instance Launch and SSH Connection

In the first couple of steps of the assignment we are to launch a new Ubuntu instance, enable SSH access, and choose the appropriate key pair settings. A best practice would be to limit SSH traffic to certain IP addresses and to review other security protocols during creation. This is followed by connection through SSH to our instance. The following shows the connection using PuTTY in the foreground and the instance detail pane in the background. This demonstrates completion of these steps.

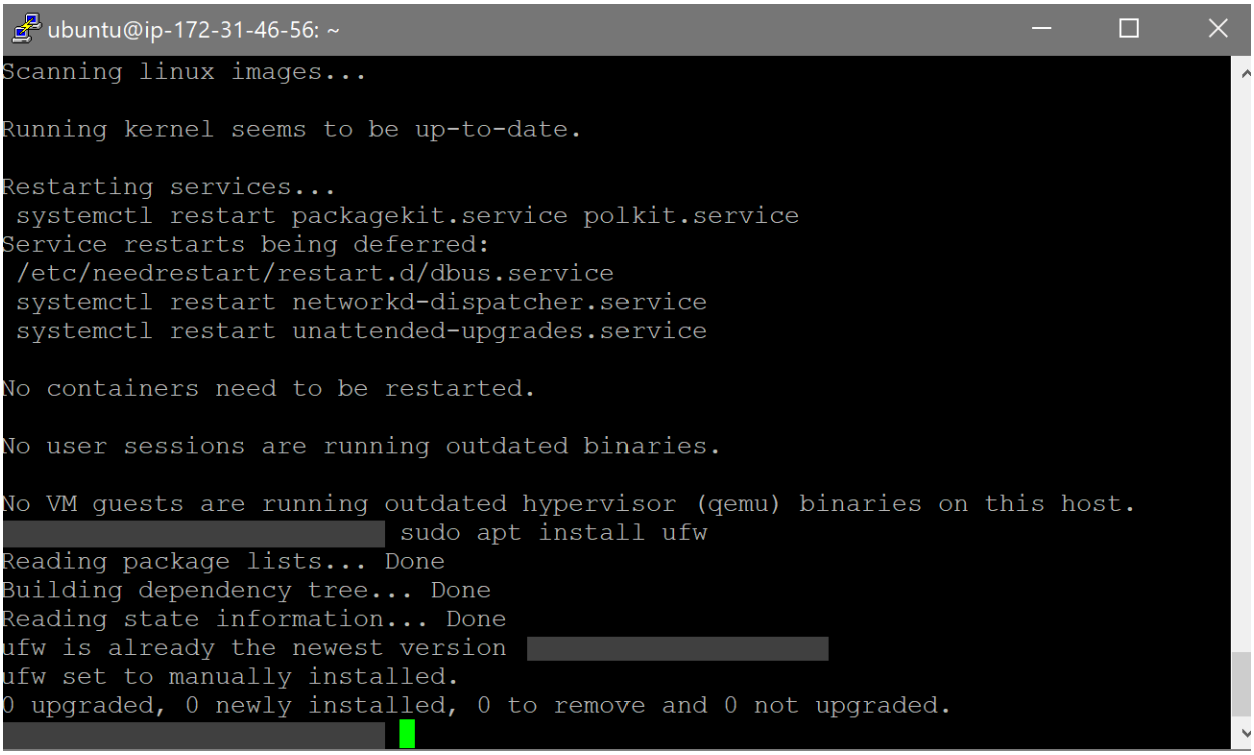


Uncomplicated Firewall

One of the first things I have been doing in the assignments so far is to update and upgrade through the package manager. The tail end of the output from this will be visible at the top of the first screenshot, below.

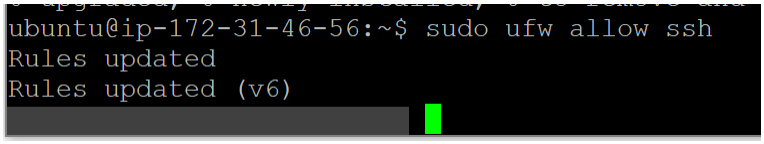
UFW Install

We use the common package manager to install UFW. In this case it was already installed and had been updated and/or upgraded with the previous commands. This is shown in the following screenshot.



SSH Access

Before enabling UFW we need to enable SSH traffic. If this is not done, our current connection can be severed, and we can be locked out. This is shown below.



Enabling the Firewall

Next, we simply enable the firewall. If we had not allowed SSH traffic in the previous step, this is when connection would be lost.

```
Rules updated (v6)
                    sudo ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y|n)? y
Firewall is active and enabled on system startup
                    
```

Status Verification

To ensure that the firewall is active and that SSH traffic is allowed, we check the status. If the firewall were active and SSH was not allowed it would be apparent without this step.

```
                    sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW       Anywhere
22/tcp (v6)                ALLOW       Anywhere (v6)

                    
```

Conclusion

This assignment is a refresher for the process of managing firewall rules and access controls. This same process was covered in the steps of the guide to setting up a VPN server in week two. It was well timed with a return from spring break and refreshed my interest in the topic of the course.

References